IT3025 Assignment 5: Coach Rowan Fitness and Nutrition Notes

Study notes - macros, calories, training basics, muscles, recovery

Macronutrients

The three macronutrients are protein, carbohydrates, and fat. Protein provides 4 calories per gram,

carbohydrates provide 4 calories per gram, and fat provides 9 calories per gram, which makes fat the

most calorie-dense macronutrient. Protein is the main nutrient used to repair and build muscle tissue.

A common guideline for people who lift weights is about 0.7 to 1 gram of protein per pound of body

weight per day. Carbohydrates are the body's preferred fuel for high-intensity exercise, and they are

stored in the muscles and liver as glycogen. Fat supports hormone production and the absorption of

vitamins A, D, E, and K.

Calories and Energy Balance

A calorie is a unit of energy. A calorie surplus means eating more calories than you burn, which

supports weight gain and muscle growth. A calorie deficit means eating fewer calories than you burn,

which leads to fat loss. Eating at maintenance means calories in roughly equal calories out, so body

weight stays the same. A rough rule of thumb is that one pound of body fat is about 3,500 calories.

Total daily energy expenditure includes your basal metabolic rate (the calories your body burns at

rest), the calories burned during exercise, and the calories burned in everyday movement.

Progressive Overload

Progressive overload means gradually increasing the demand placed on the muscles over time. You can

apply it by adding weight, adding reps, adding sets, or reducing rest time. Progressive overload is

the main reason muscles get stronger and bigger, because the body only adapts when it is challenged

beyond what it is used to. Lifting the same weight for the same reps forever leads to a plateau.

Reps, Sets, and Rest

A rep (repetition) is one full completion of an exercise movement, and a set is a group of

consecutive reps. Lower rep ranges of about 1 to 5 reps per set with heavy weight build maximal

strength. Moderate rep ranges of about 6 to 12 reps per set are most associated with muscle growth,

also called hypertrophy. Higher rep ranges of 15 or more reps per set build muscular endurance. Rest

between sets is usually about 2 to 3 minutes for heavy compound lifts and about 60 to 90 seconds for

lighter accessory work.

Compound Versus Isolation Exercises

A compound exercise works several muscle groups and joints at once, such as the squat, bench press,

deadlift, and row. An isolation exercise targets one muscle and one joint, such as the bicep curl or

leg extension. Compound lifts are usually the foundation of a program because they train more muscle

in less time.

Muscle Groups and Key Lifts

The squat mainly works the quadriceps and glutes. The bench press mainly works the pectoralis major

(chest), the triceps, and the front of the shoulders. The deadlift mainly works the hamstrings, the

glutes, and the lower back (erector spinae). The barbell row mainly works the latissimus dorsi

(lats), the rhomboids in the upper back, and the biceps. The overhead press mainly works the

deltoids (shoulders) and the triceps.

Recovery

Muscles grow during recovery, not during the workout itself. Training creates tiny tears in muscle

fibers, and the body repairs them stronger during rest, fueled by protein and sleep. Most adults

should aim for 7 to 9 hours of sleep per night. A common guideline is to wait about 48 hours before

training the same muscle group hard again. Skipping rest days and sleeping too little can stall

progress and raise injury risk. Staying hydrated also supports performance and recovery.

Warm-Up and Form

A warm-up raises body temperature and prepares the joints and muscles for heavier work, and it

usually includes light cardio and lighter practice sets of the exercise. Good form means controlling

the weight through a full range of motion. Using too much weight with poor form is a common cause of

injury.